

KOTKA HAAPASAARI, Finland

WMO: 029670

Lat: 60.287N

Lon: 27.185E

Elev: 6

StdP: 101.26

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.4	-15.7	-20.9	0.6	-18.1	-18.1	0.8	-15.6	16.3	-2.1	15.1	-1.9	4.9	20	0.779

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.0	24.5	20.3	23.1	19.4	21.8	18.5	21.1	23.8	20.0	22.9	18.9	21.4	4.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	14.7	23.2	18.9	13.7	22.3	17.7	12.7	20.8	61.1	23.8	57.4	23.0	53.6	21.5	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 15.2	13.6	12.3		-18.6	26.2	5.2	2.8	-22.3	28.2	-25.4	29.9	-28.3	31.4	-32.1	33.4	
(5)			DB	-18.9	21.3	5.1	1.4	-22.6	22.3	-25.6	23.1	-28.4	23.9	-32.2	24.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-3.6	-4.9	-2.2	3.1	9.5	14.0	18.2	17.7	13.5	7.9	3.5	0.4
	DBStd	8.67	5.27	5.67	4.25	2.49	3.53	2.78	2.94	2.44	2.54	3.05	3.44	3.91
	HDD10.0	2056	422	416	379	207	53	2	0	0	4	79	195	297
	HDD18.3	4381	681	649	638	457	275	133	39	41	144	324	445	555
	CDD10.0	776	0	0	0	1	37	122	255	238	111	13	0	0
	CDD18.3	60	0	0	0	0	0	3	36	21	0	0	0	0
	CDH23.3	104	0	0	0	0	0	6	79	19	0	0	0	0
	CDH26.7	5	0	0	0	0	0	1	4	0	0	0	0	0
(14) Wind	WSAvg	6.5	7.3	6.3	6.3	5.6	5.4	5.7	5.2	5.7	6.6	7.3	8.0	8.4
(15) Precipitation	PrecAvg	665	53	44	38	32	41	57	64	74	64	71	68	59
	PrecMax	914	101	108	79	71	114	124	113	155	141	164	156	147
	PrecMin	492	6	2	1	6	9	11	4	8	14	17	7	15
	PrecStd	99	22	25	19	16	27	24	30	36	38	38	31	26
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	3.2	5.2	13.0	20.8	23.7	26.9	25.2	19.6	14.2	10.0	7.7
		MCWB	4.0	1.9	2.6	8.5	14.2	17.8	21.1	21.0	16.7	12.8	9.0	6.9
	2%	DB	4.0	2.2	3.7	9.3	18.5	21.4	25.3	23.6	18.2	13.5	9.2	6.6
		MCWB	3.4	1.5	1.9	6.0	13.2	16.9	20.8	19.9	15.9	12.3	8.4	5.7
	5%	DB	3.2	1.7	2.9	7.7	16.6	19.3	24.0	22.5	17.4	12.9	8.5	5.7
		MCWB	2.6	1.1	1.6	5.0	12.7	15.4	20.2	19.0	15.3	11.7	7.7	5.0
	10%	DB	2.4	1.1	2.1	6.4	14.5	17.8	22.6	21.2	16.7	12.0	7.6	4.9
		MCWB	1.8	0.6	1.2	4.2	11.3	14.6	19.4	18.1	14.8	10.8	6.8	4.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	2.3	3.3	8.7	15.6	19.4	22.6	21.7	17.6	13.3	9.2	7.1
		MCDB	4.6	2.6	4.4	12.5	19.0	22.9	25.3	24.5	19.0	13.8	9.6	7.6
	2%	WB	3.5	1.8	2.3	6.7	14.1	17.5	21.6	20.6	16.7	12.6	8.6	5.9
		MCDB	3.9	2.1	3.2	9.0	17.4	20.3	24.3	23.3	17.8	13.3	9.0	6.4
	5%	WB	2.7	1.2	1.9	5.4	12.9	16.2	20.7	19.6	15.9	11.9	7.8	5.1
		MCDB	3.2	1.6	2.6	7.1	16.0	18.8	23.5	22.1	17.1	12.7	8.3	5.6
	10%	WB	1.9	0.7	1.4	4.5	11.6	15.0	19.7	18.5	15.2	11.0	6.9	4.2
		MCDB	2.4	1.1	2.0	6.0	14.5	17.4	22.6	20.7	16.6	12.0	7.5	4.8
(35) Mean Daily Temperature Range	5% DB	MDBR	3.6	3.9	4.7	4.1	4.8	4.2	4.0	3.4	3.0	2.8	2.6	3.0
		MCDBR	2.7	2.5	3.7	5.9	7.5	6.3	5.3	4.3	3.3	2.6	2.7	2.7
	5% WB	MCWBR	2.8	2.3	2.7	3.4	4.2	3.4	2.8	2.7	2.5	2.6	2.8	2.7
		MCDWR	2.7	2.4	3.2	5.2	6.9	5.6	4.8	3.6	2.8	2.5	2.7	2.7
(40) Clear-Sky Solar Irradiance	taub		0.279	0.311	0.333	0.371	0.369	0.366	0.388	0.381	0.354	0.335	0.307	0.386
		taud	2.137	2.156	2.199	2.274	2.405	2.462	2.419	2.435	2.490	2.386	2.208	2.602
	Ebn at Noon		468	664	783	823	857	866	833	810	772	650	431	328
		Edh at Noon	43	74	98	110	104	100	102	94	75	60	40	29
(44) All-Sky Solar Radiation	RadAvg		0.25	0.86	2.34	3.87	5.31	5.52	5.51	4.21	2.50	1.04	0.28	0.14
	RadStd		0.05	0.18	0.30	0.42	0.57	0.38	0.49	0.43	0.29	0.15	0.05	0.02

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)		+0.49	N/A	N/A	N/A	N/A	N/A	-168	-177	N/A	N/A

Nomenclature: See separate page